REN AXIS

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Urban Design for the Centennial Jing-Zhang AI Innovation Belt (AI-generated concept)

submissions/Abreto/ren-axis-jingzhang-ai-belt · v5.8 · 2026-08-11

Contents · Version · Team

A3 booklet (Chinese 20 pp / English 18 pp)

Booklet structure

Generated by

AI agent: Claude Fable 5
Operator GitHub: Abreto
Method: deterministic scripts derive geometry from the provisional boundary and recompute metrics in EPSG:4548; text and drawings share one source
Disclosure: agent.json · self_check.json

Version and licence

Version v5.8 · 2026–08–11 (fully logged in changelog.md)

PDF text is Type 3 vector outline (zero /FontFile entries, no installable font embedded)
Provisional boundary: recompute in full once official geometry arrives

Evidence guide for reviewers

Metrics: per–item formula fields in metrics.json
Topology: EPSG:4548 review, matching the spatial check digit for digit
Governance ledgers: shipped as JSON and shown in the dashboard
Rights ledger: report/copyright_statement.md, registered by class

Design Basis and Source List

All sources registered in sources.json and cited in the text

Task basis

Pre–qualification announcement for the international call on the Centennial Jing–Zhang AI Innovation Belt (2026–05–09)
Open–call task book for global agents (2026–05–18)

Professional standards

Urban Design Administration Measures (MOHURD)
Measures for Regulatory Detailed Plans (MOHURD)
Land and Sea Use Classification Guidelines (MNR)
Architectural depth rules: pending official text

Data base

Site package brief/site–package
Public source registry source_registry.json
Processed fact pack data/processed
Provisional boundary provisional_boundaries.geojson

Boundary status

The official precise red line is missing; every layer is generated on a provisional rough boundary
Not valid as an official red line, approval basis or precise area; recompute in full once polygons arrive

Three-Level Scope Framework

Research → overall design → key areas, level by level

Coordinated research area

About 43.6 km²: Fifth Ring — Jingzang Expressway — Xizhimenwai Street — Wanquanhe Road
Task: industrial strategy, future urban form, regional synergy

Overall design area

About 11.4 km² (recomputed 11.41 km²)
The 1—2 km urban band around the rail-heritage park
Task: regulatory-depth urban design and renewal framework

Key area scope

About 368.4 ha (recomputed 369.3 ha)
Zhongzhiyuan 192.9 / Origin 104.3 / Dazhongsi 72.0 ha
Task: detailed design at implementation-plan depth

Transmission logic

Strategy sets direction (ecosystem, loop, brand)
Design sets structure (axis, stations, wings, land use)
Implementation sets projects (three stations, 18 projects)

Existing conditions (method framework)

Based on public material; parcel-level diagnosis awaits official survey data

Severance

Cut east-west by the Third Ring, Zhichun Road and
the Fourth Ring; the park's slow-mobility system breaks
The announcement explicitly asks for innovative solutions

Underused space

Underused buildings and legacy trade facilities in the belt
Renewal potential is scattered and needs unit coordination
Dazhongsi shows the strongest need for business conversion

Weak integration

Campus, park and neighbourhood boundaries are separated
Innovation factors circulate poorly
The live-work-service mix needs rebalancing

Data gaps

Existing buildings, tenure, regulatory conditions,
heritage lines and municipal capacity are all unverified
This is a method framework, not parcel-level conclusions

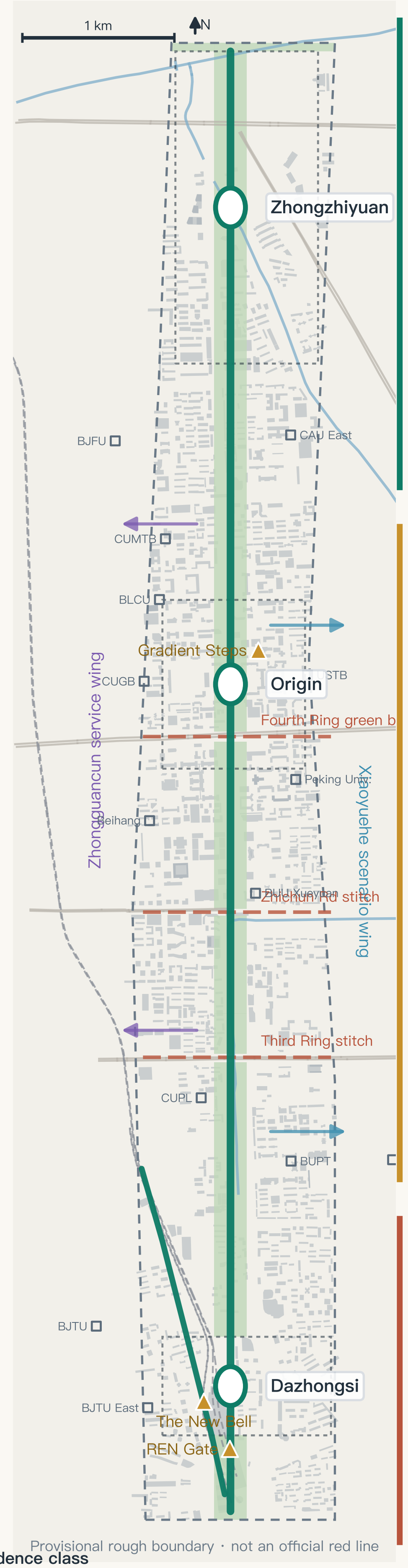
Giving the Character 人 Back to Jing–Zhang

REN AXIS

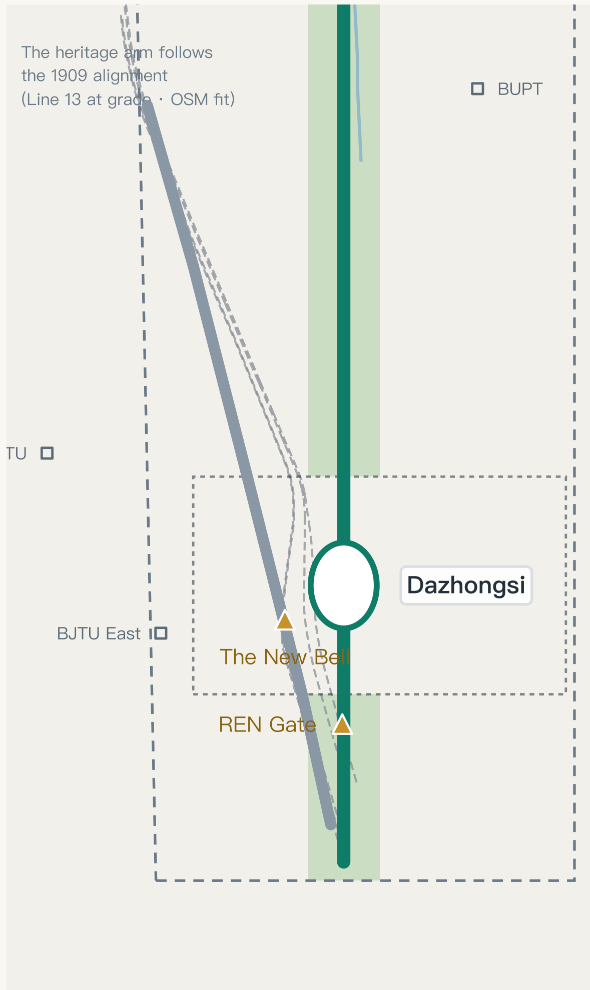
Centennial Jing–Zhang AI Innovation Belt · AI–generated concept proposal

One axis, two arms, three stations, two wings — the future and heritage arms meet at the REN Gate

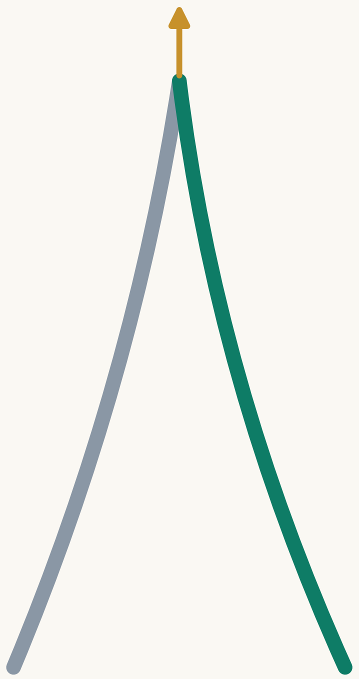
Overall spatial structure (on provisional boundary)



The fork, close up: future arm (green) × heritage arm (silver)



Concept prototype



Logo direction: two rails converge into 人
reading the two alignments × gradient ascent

Visual identity (concept)

- Signal green
- Rail silver
- Quantum blue
- Rammed–earth gold

Wordmark: REN AXIS

*Trademark search and clearance required before formal use

Gradient narrative: a climb one century long

Summit · north

Liudaokou — Qinghe green wedge

Buildings step back · greenest
Marshalling–yard prototype inside a garden
The axis closes at the green wedge

Switchback · middle

Third Ring — Xueyuan Road

Three stitching corridors as three switch throws
Scene density rises along the axis
The heritage arm unfolds the old line

Departure · south

REN Gate — Dazhongsi

City gateway · tallest interfaces
The two arms fork at Switch Plaza
Densest programme · station–city living room

Interface height, scene intensity and landmarks climb together
The one–day route is its public version

Naming system (concept)

Main name: REN AXIS · Centennial Jing–Zhang AI Innovation Belt
Sub–systems: three stations · switch nodes ·
signal honours · K–markers
*Trademark search required before formal use

Three positions × five functions

Century cultural belt → K–markers · landmarks · station memory
Urban AI life belt → Xiaoyuehe wing · 12 scenarios
AI convergence belt → three–station, two–wing ecosystem
Full–stack → Zhongzhiyuan · Ecosystem → Origin
AI–native business → Dazhongsi

Collaboration loop

Origin (research) → Zhongzhiyuan (engineering)
→ Dazhongsi (product) → Zhongguancun (capital)
→ Xiaoyuehe (validation) → back to research
Regional: Future Science City · Huairou ·
Economic Development Area · Jing–Jin–Ji

Evidence class

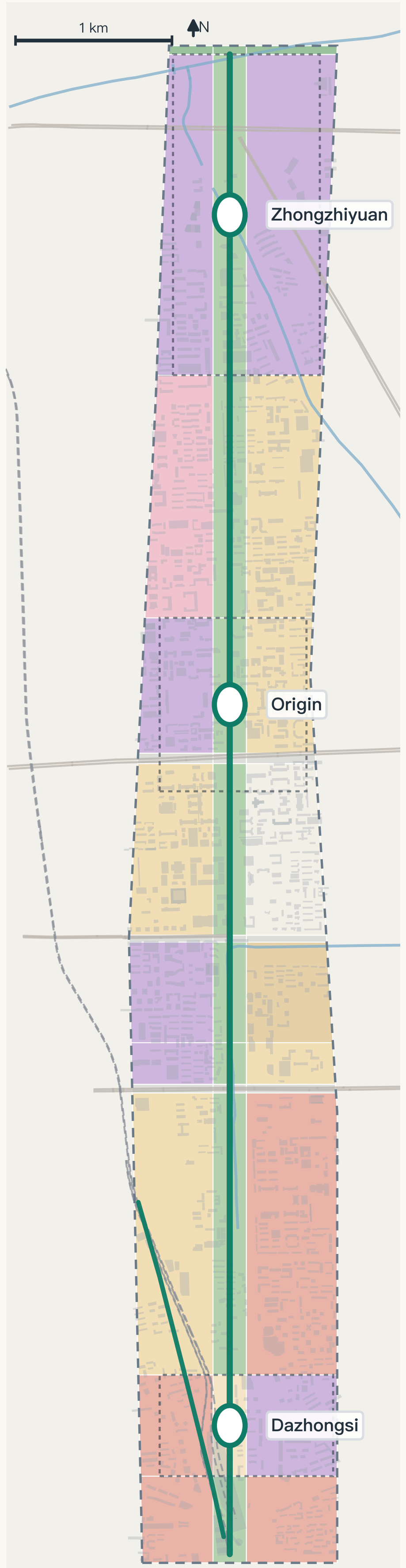
- official fact · announcement figures and tasks
- OSM concept alignment · existing alignments and points (concept level)
- provisional proposal · this proposal's spatial suggestions (provisional)

Data: submissions/Abreto/ren–axis–jingzhang–ai–belt geometry/*.geojson and metrics.json (recomputed in EPSG:4548) · Existing features © OpenStreetMap contributors (ODbL) · Boundary is a provisional substitute; not valid as an official red line or precise area basis

Three scope levels, one transmission; thirty-nine land-use cells in one pass

Research 43.6 km² → design 11.4 km² → key areas 368.4 ha; thirty-nine cells, complete division, no overlaps, gap below tolerance

Land use (concept · legend right)



Provisional rough boundary · not an official red line

Nested scopes (recomputed in EPSG:4548; Zhongzhiyuan 192.9 · Origin 104.3 · Dazhongsi 72.0 ha)

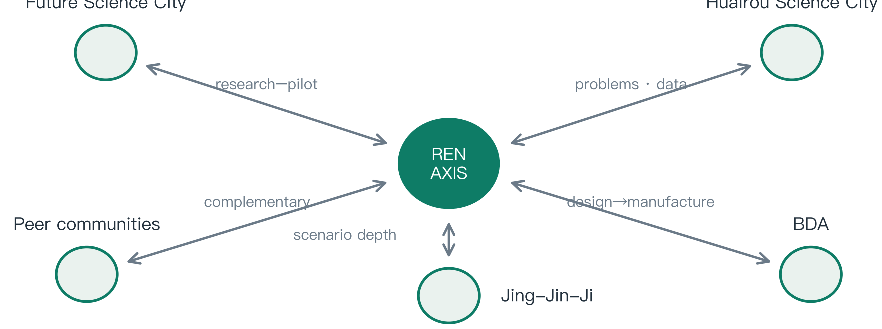
Three scales

Structure notes

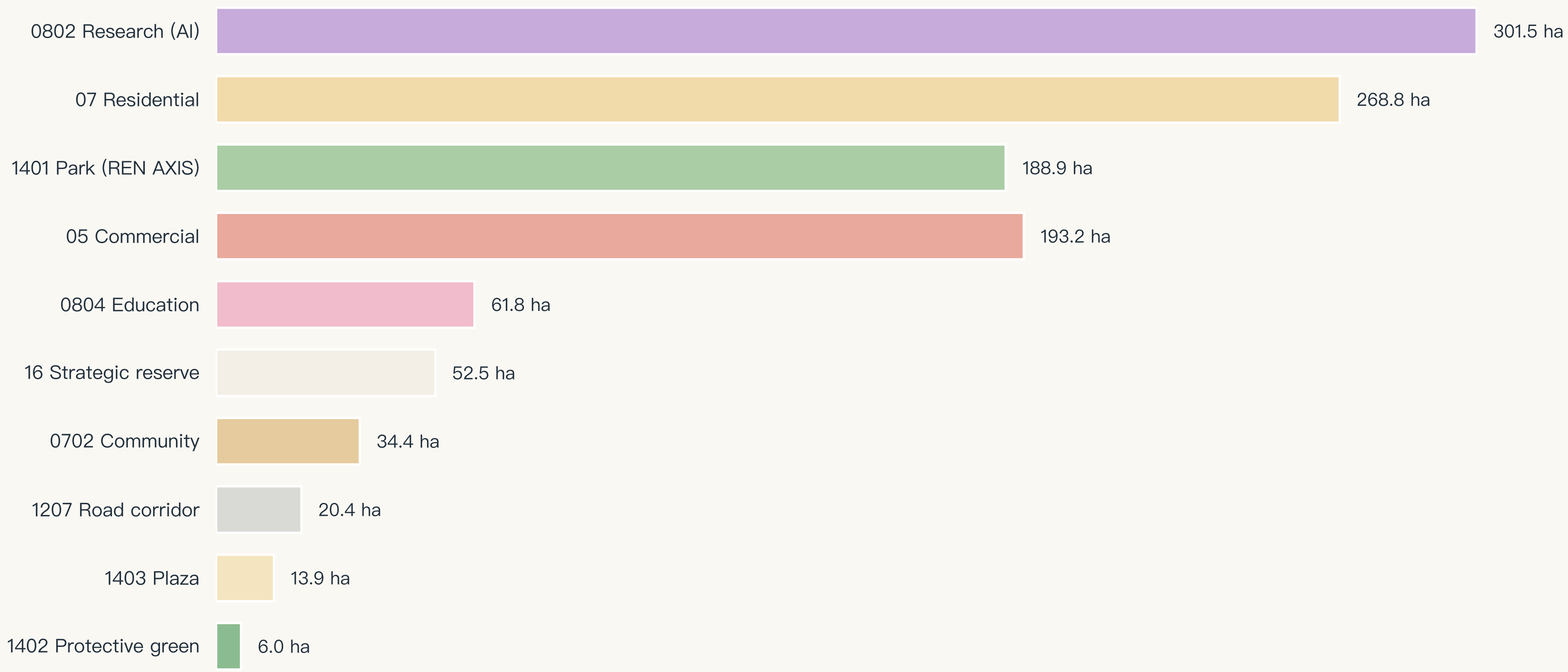
- Research land 301.5 ha, the largest class: the AI industrial backbone
- Park land 188.9 ha runs the full length of the axis
- Residential + community 303.2 ha supports live-work balance
- Three road corridors form the east-west stitching interfaces
- Reserve 52.5 ha holds long-term flexibility

*All concept suggestions, subject to the statutory plan

Regional interfaces (concept, not commitments)



Land-use area by class (hectares, recomputed from land_use.geojson)

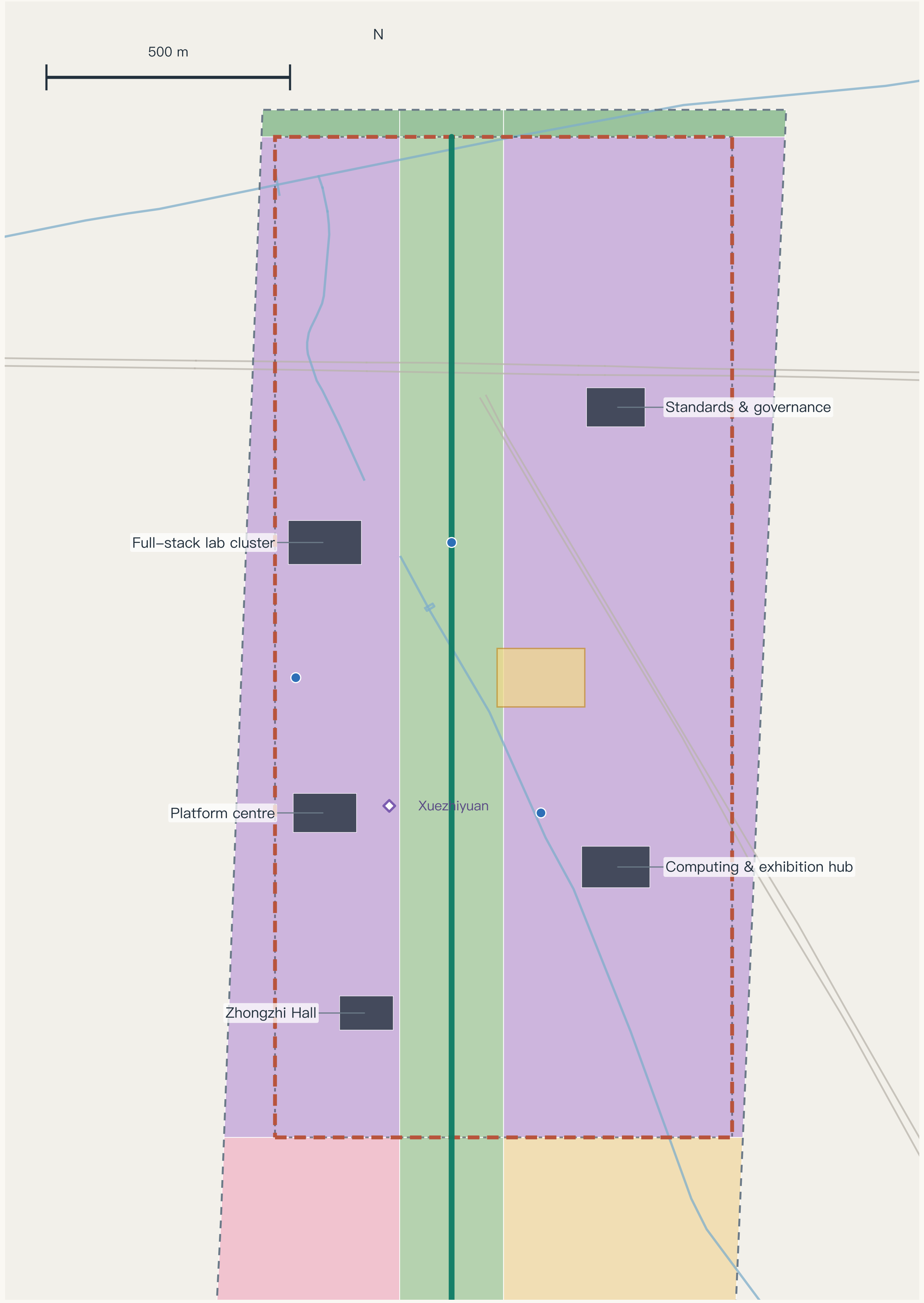


Evidence class

- official fact · announcement figures and tasks
- OSM concept alignment · existing alignments and points (concept level)
- provisional proposal · this proposal's spatial suggestions (provisional)

Key Area Detail · Zhongzhiyuan Acceleration Terminal

Garden–type innovation district · 192.9 ha (recomputed; key–area polygon is provisional)



Positioning and systems

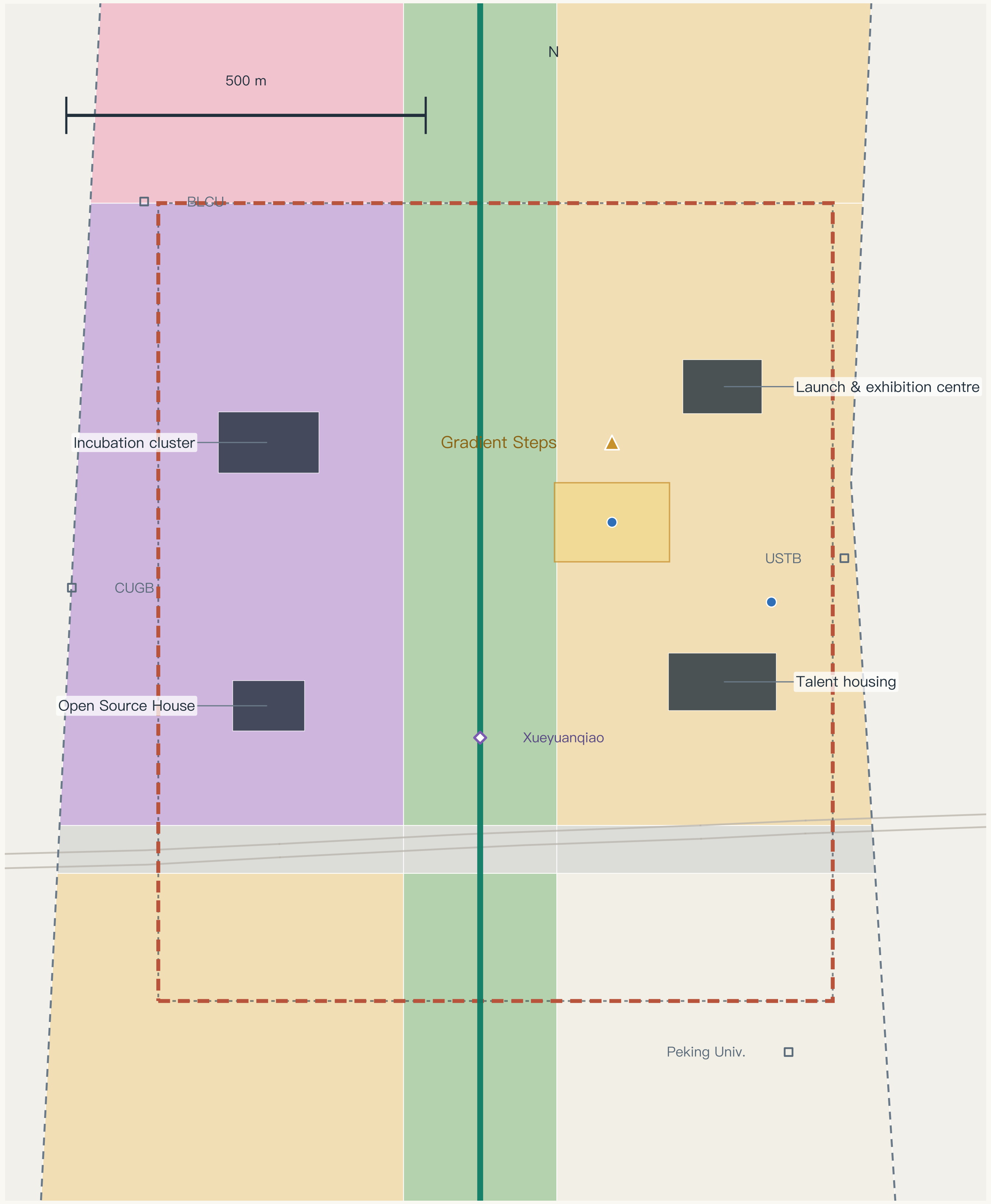
Full–stack innovation / AI governance voice
Lab clusters · governance centre · computing hub · Zhongzhi Hall
Innovation loop + ring–road integration study
Risks: tenure · ecology–intensity balance
Operations: scenario list system; time–boxed loop testing

Common note

Morphology · block scale · ground floor · public space · risk
The five–line template is set out in chapter five
All conclusions are concept suggestions and references
for professional teams to develop
Retain–renovate–demolish is a method framework only

Key Area Detail · Origin Source Station

Campus–adjacent district · 104.3 ha (recomputed; key–area polygon is provisional)



Positioning and systems

World–class AI innovation ecosystem
Incubation cluster · Open Source House · talent housing
Campus gallery + station integration study
Risks: campus coordination · low disturbance
Operations: permanent Open Source House; regular launch days

Common note

Morphology · block scale · ground floor · public space · risk
The five–line template is set out in chapter five
All conclusions are concept suggestions and references
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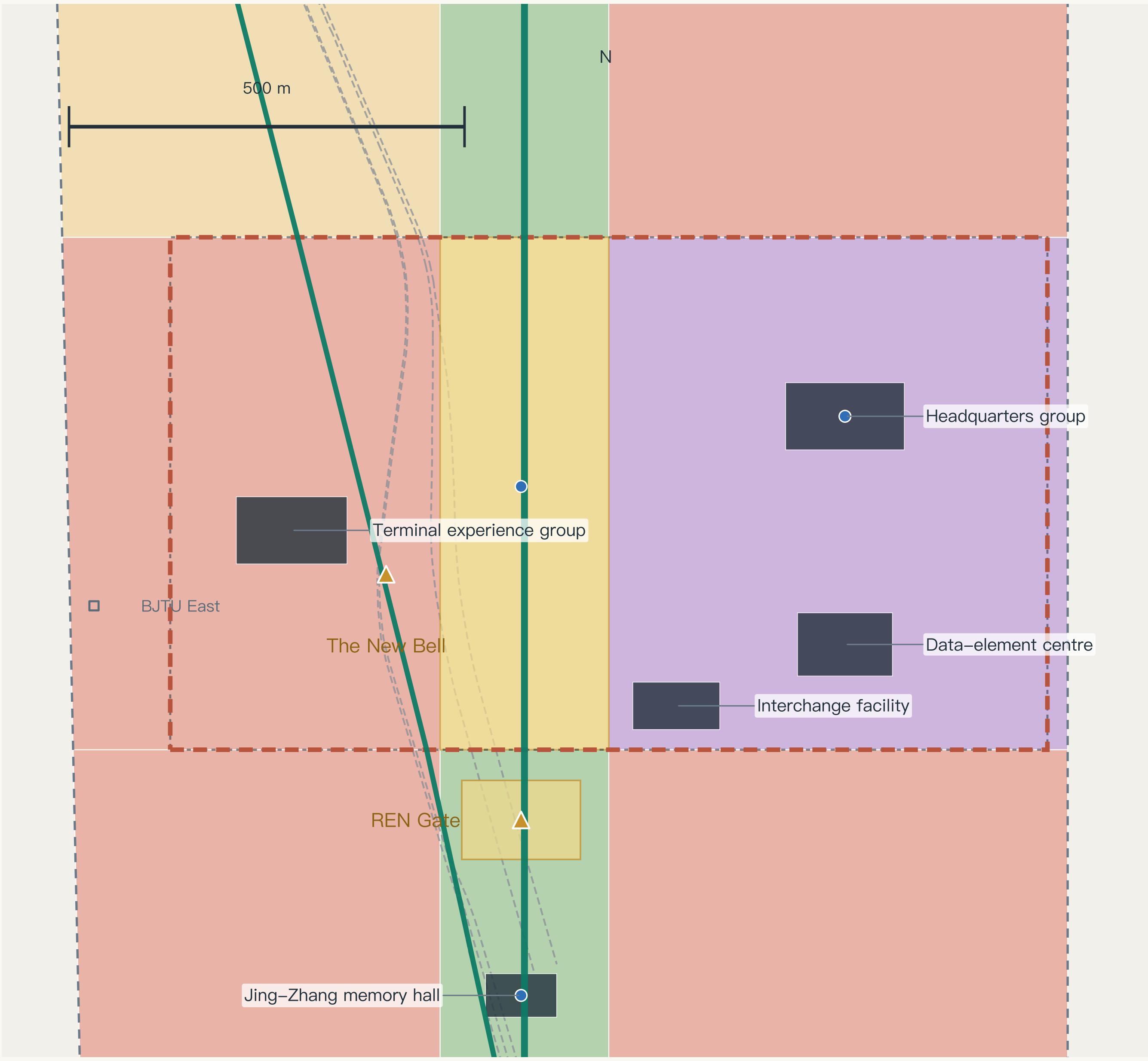
Key Area Detail · Dazhongsi Exchange Terminal

Urban–type innovation district · 72.0 ha (recomputed; key–area polygon is provisional)



Positioning and systems

AI–native business formats
Consumption showcase · headquarters · data–element centre
Four–quadrant pedestrian connection at the forecourt
Risks: heritage constraints · traffic pressure
Operations: showcase calendar; time–shared forecourt

> This rectangle has no station or road anchoring yet — hence the REN Gate and the Jing–Zhang memory hall falling inside it, and a centroid about 2.26 km from Dazhongsi station. The position awaits regeneration once official anchoring is in place.

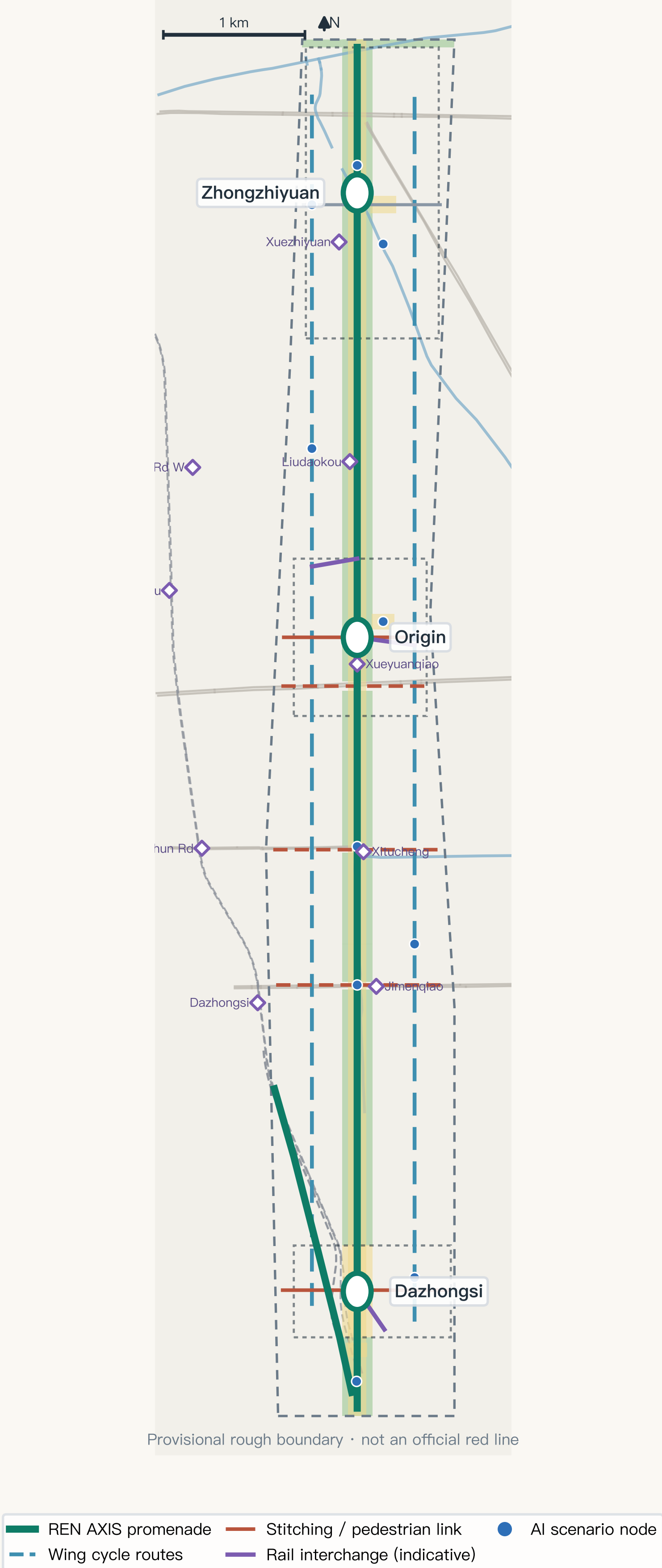
Common note

Morphology · block scale · ground floor · public space · risk
The five–line template is set out in chapter five
All conclusions are concept suggestions and references
for professional teams to develop
Retain–renovate–demolish is a method framework only

36.6 km of slow-mobility network, stitching a severed belt back into the city

Slow-mobility and interchange network 36.6 km · green ratio 17.1% · public space ratio 11.8% (includes the axis band)

Slow mobility, interchange and blue-green (concept alignments; ◇ existing rail stations, OSM)



Provisional rough boundary · not an official red line

REN AXIS route diagram (south → north)

Three stitching strategies (concept)

Option ① Green bridge: overpass link study at Fourth Ring / Xueyuanqiao
Option ② Underpass: improvement study at the Third Ring node
Option ③ At-grade crossing: signal timing study at Zhichun Road
*Three parallel options, compared per corridor; selection requires professional appraisal

Station-city integration (research suggestion)

Dazhongsi: four-quadrant pedestrian link + interchange
Toward Wudaokou / Tsinghua East Road West Gate: connection
Station positions indicate the public network only
Cycles: stacked parking organised at the forecourt

Continuous blue-green system

Green space 194.9 ha (17.1%)
Meets Qinghe in the north · answers Xiaoyuehe in the east
Public space 134.2 ha: four plazas + promenade activity band
Edge computing pods share poles with smart posts along the axis

Evidence class

■ official fact · announcement figures and tasks
■ OSM concept alignment · existing alignments and points (concept level)
■ provisional proposal · this proposal's spatial suggestions (provisional)

Data: submissions/Abreto/ren-axis-jingzhang-ai-belt geometry/*.geojson and metrics.json (recomputed in EPSG:4548) · Existing features © OpenStreetMap contributors (ODbL) · Boundary is a provisional substitute; not valid as an official red line or precise area basis

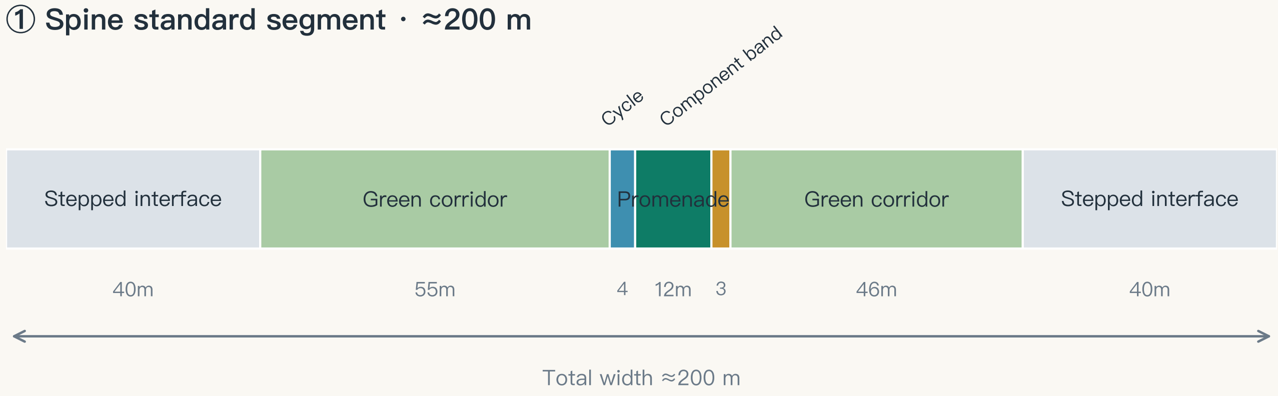
Six Street Sections (conceptual study dimensions)

REN AXIS

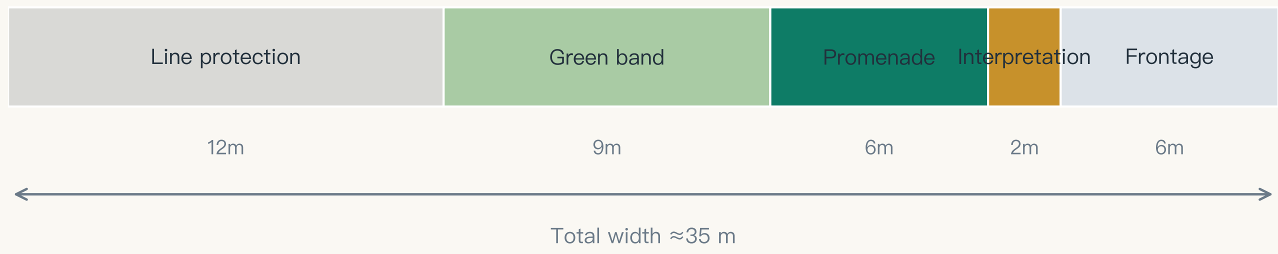
Centennial Jing–Zhang AI Innovation Belt · AI–generated concept proposal

Spine · heritage arm · ring stitch · campus interface · station street · waterfront — study sections, not engineering drawings

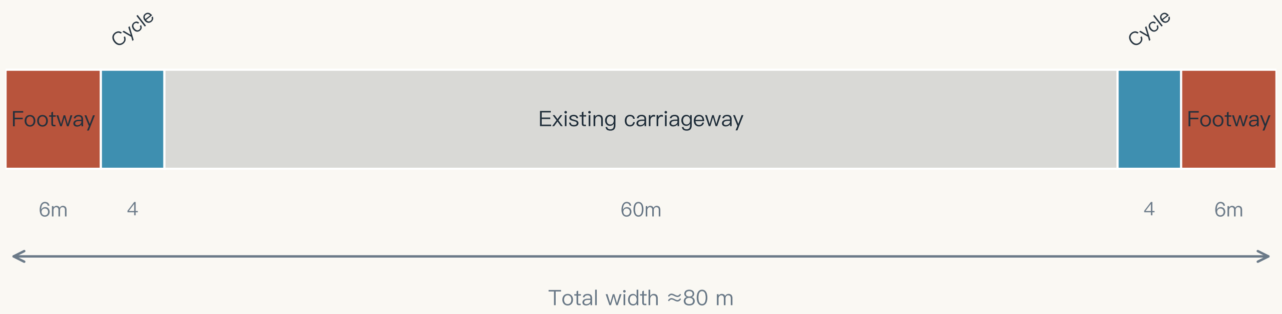
① Spine standard segment · ≈200 m



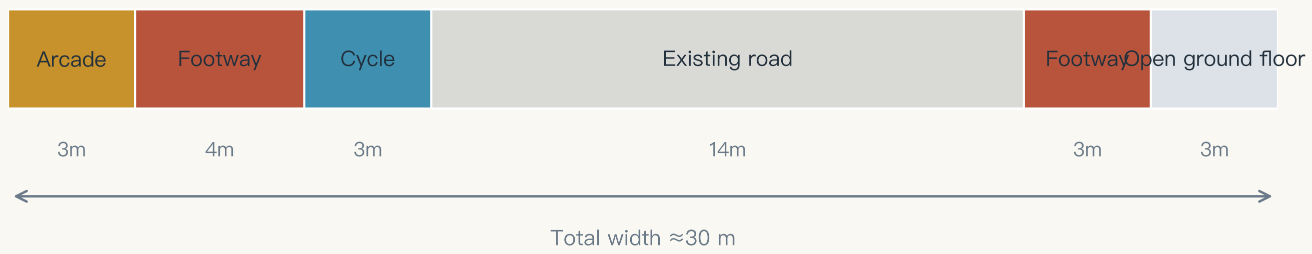
② Heritage arm segment · ≈35 m



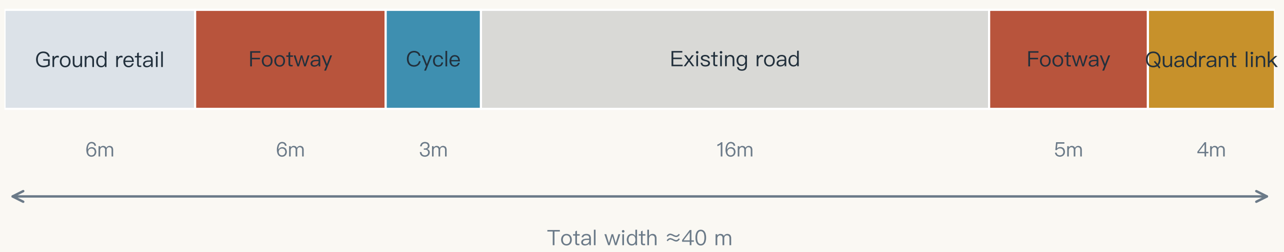
③ Ring stitch (Third Ring) · ≈80 m



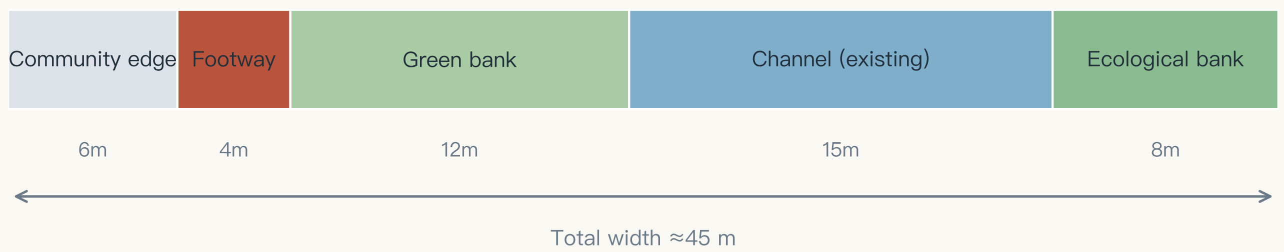
④ Campus interface street · ≈30 m



⑤ Station–front street (Dazhongsi) · ≈40 m



⑥ Xiaoyuehe waterfront · ≈45 m



Sections follow the gradient narrative: tallest interfaces at departure, densest stitching at the switchback, greatest green at the summit; promenade and green corridor run continuously.

AI Scenarios and Long–Term Operations

Twelve scenario cards (three test–and–validation) · annual programme · community operations

Scenario cards (south)

SC–04 AI interpretation and cultural narrative (REN Gate)
SC–10 Night–time vitality and safety care (Dazhongsi)
SC–06 Enterprise service copilot station
SC–01 Axis commuting assistant (Third Ring)

Scenario cards (middle and north)

SC–02 Accessible mobility / SC–07 Health navigation
SC–05 Open–source market / SC–03 Robot delivery (test)
SC–08 Public AI learning pod / SC–09 Eco monitoring (test)
SC–11 Autonomous shuttle (test) / SC–12 Urban model ground

Annual programme

Spring · global developers' conference (Zhongzhiyuan forecourt)
Summer · open–source summer camp (Open Source House)
Autumn · AI art and city festival (whole axis)
Winter · model release season and honours night (Gradient Steps)

Operations and conversion

Developer community: member co–creation, mentor pairing, corporate challenges
Scenario opening: publish, apply, review, test, human check
Conversion: events → talent station → incubation → space supply
*All concept suggestions, not confirmed government arrangements

Executable Pilot Cards · SC–01 / SC–09 / SC–12

Eleven fields each: what must be measured before a pilot may start, who benefits and bears risk, what it needs, and who may stop it and how to appeal

SC–01 Axis commuting assistant

Third Ring stitching segment

Baseline

Four weeks of pre–intervention crossing wait times and footway flows at the same points and hours; frozen from a measured baseline before the pilot; no value stated yet

Trial window and sample

12 weeks; two peak hours on weekdays plus one weekend window; sample is section–level aggregate counts with no individual identifiers

Minimum success threshold

Median crossing wait improves against baseline with zero safety incidents; the improvement threshold is frozen from a measured baseline before the pilot; no value stated yet

Stop threshold

Any safety incident, or advice inconsistent with the actual signal state reaching the set count, or a complaint unanswered within one working day — red card, revert to conventional crossing infrastructure

On–site owner

The operator's traffic watch team (a role, not a named body); community representatives hold an equal red–card right

Human equivalent service

Conventional crossings and the original signal timing stay fully available throughout; the pilot never degrades the baseline service

Proof of data deletion

Aggregate counts kept ≤90 days then auto–deleted; on red card or retirement the cache is cleared immediately and a deletion record is issued (time, scope, executor, verifier)

Review cycle

Review every four weeks with published minutes; after a red card, accountability review and affected–group feedback must complete before a yellow observation period

Who benefits and who bears risk

Benefits: crossers on both sides of the North Third Ring, including people with accessibility needs. Risk bearers: the same crossers (misleading guidance) and the watch team (response burden). Equal service, no vulnerable groups in round one, affected–group red–card power

Resources and permissions

Needs: read–only signal–timing data (timing authority unchanged), installation permission and power for the counters, a spot for the QR code; permitting bodies are confirmed at the detailing stage — no permission, no start

Appeal and accountability path

A review role independent of the watch team confirms the sample definition; deletion records go to that role and the community representatives, readable in the opinion—response ledger; appeals enter via QR code or the public channel — an objection that triggers no card can still demand one review and gets a three–state answer

SC–09 Smart horticulture and ecological monitoring (test)

Zhongzhiyuan green corridor

Baseline

A first–season human control group: recognition accuracy and water use under conventional scheduled maintenance; frozen from a measured baseline before the pilot; no value stated yet

Trial window and sample

One full growing season; sample is fixed–position vegetation imagery and environmental sensing with no people in frame; a matched control section runs alongside

Minimum success threshold

Both recognition accuracy and water saving beat the human control group; both thresholds are frozen from a measured baseline before the pilot; no value stated yet

Stop threshold

Any erroneous action endangering vegetation or the public triggers a red card; two consecutive periods below target trigger a yellow card

On–site owner

Landscape maintenance operator and technology supplier jointly named (roles); the maintenance crew holds on–site stop authority

Human equivalent service

Conventional scheduled maintenance can be reinstated at any time, and reinstatement does not require the supplier's consent

Proof of data deletion

Raw imagery kept ≤180 days; on retirement it is deleted under the agreed rule with a deletion record issued, while evaluation results and the failure log remain public

Review cycle

Monthly inspection plus quarterly controlled evaluation; every failure is logged publicly so later scenarios can avoid the same trap

Who benefits and who bears risk

Benefits: the corridor's vegetation and its daily users (better maintenance, less water). Risk bearers: the vegetation itself (mis–irrigation, mis–pruning) and the maintenance crew (workload). The trial tests vegetation, not the public; imagery contains no people

Resources and permissions

Needs: the maintenance work permit, consent for fixed camera positions, controlled access to the irrigation interface; water and power run under the existing maintenance arrangement, bodies confirmed at the detailing stage

Appeal and accountability path

A maintenance review role independent of the supplier confirms the control group's sections and indicators; deletion records go to that role, readable in the ledger; the notice board carries the feedback entry — an objection that triggers no card can still demand one review with a three–state answer

SC–12 Urban model ground

Zhongzhiyuan exhibition hub

Baseline

This package's current metrics are the baseline: 50 recomputed metrics and 4 control values pending official data, as declared in metrics.json

Trial window and sample

Rolls with each version K–marker; every arrival of official data triggers one whole–package recomputation and display update

Minimum success threshold

Displayed values match metrics.json digit for digit at 100%, and pending metrics never appear as numbers in the display layer — an immediately enforceable rule that needs no pilot

Stop threshold

Any layer whose values disagree is withdrawn with a published reason; if the build–time assertion fails, that version is not released

On–site owner

Exhibition operator and a data–definition committee (roles); the public appeal route is an issue in the open–source repository

Human equivalent service

Static drawings and a printed metrics handbook are always available; loss of power or network does not affect access

Proof of data deletion

Only public datasets are used and no personal data is ingested, so no personal–data deletion obligation arises; on withdrawal the exhibit is archived

Review cycle

A human consistency check after every version K–marker update, with the check record retained

Who benefits and who bears risk

Benefits: the public consulting the model and later agents (a recomputable public data asset). The risk is cognitive — an inconsistent display would mislead readers, which is why consistency is the success criterion. No personal data, so no individual bears data risk

Resources and permissions

Needs: exhibition space and power at the hub (sited with the phase–one exhibition–hub project), open–source hosting (in place); only public datasets, no personal–data permission needed

Appeal and accountability path

Consistency is confirmed twice — build–time assertion in the pipeline, recorded human check; the repository issue channel is the public appeal route: anyone can challenge an inconsistency, gets a three–state answer, and reads the handling and any withdrawal reason in the ledger

These three cards answer comment 002 in the public channel (issue #215); thresholds come from a measured pre–pilot baseline and are never invented — SC–12 excepted, its criterion being internal consistency, enforceable now. Responses 003—004 (2026–08–10) added the benefit/risk, resources–and–permissions and appeal–and–accountability fields; the appeal field answers comment 003's three gaps — an independent confirmer for the sample definition, whom deletion records go to and where to read them back, and a review that external objections can trigger on their own.

Governance Ledgers (machine–audit layer)

Full field–by–field records ship as language–neutral JSON with the package

Scenario governance ledger

Twelve scenario cards × thirteen fields:
responsible role · tenure and permit preconditions ·
minimum data · privacy and retention · legal interface ·
KPI baseline and threshold · human review · acceptance ·
non–AI equivalent path · suspend · appeal · recover · retire
→ visual/assets/governance–ledger.json

Implementation ledger

Eighteen renewal projects × nine fields:
responsibility type · phase · investment band ·
tenure status · preconditions · demand basis ·
human review · acceptance evidence · suspend and exit
→ visual/assets/governance–ledger.json

Agent operating protocol

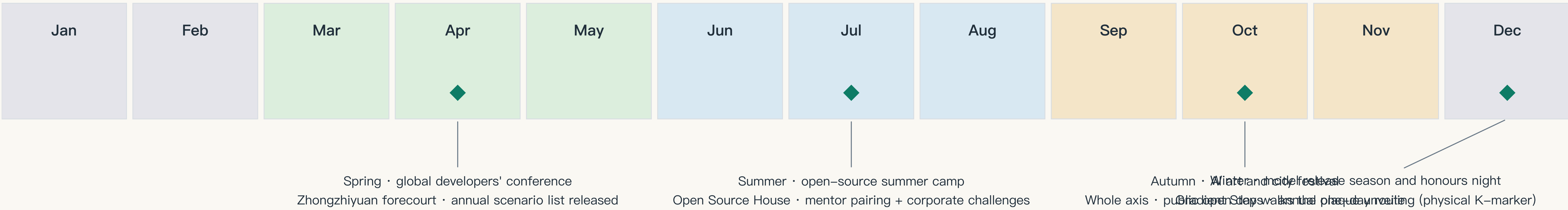
State machine formalised: seven states, eight transitions, human sign–off points, and the red card as the only immediate transition needing no prior sign–off.
Per–scenario agent interface: what it may read, decide, must escalate, and how it is halted.
→ visual/assets/agent–protocol.json

Why they are not printed here

The proposal format separates a human reading layer from a machine audit layer. These ledgers are exhaustive records built for validation, not for reading at A3 size; the Chinese booklet prints them for domestic review, and both versions resolve to the same JSON source.

Annual Operating Calendar · Long–Term Mechanism (agent.6)

Four seasonal flagships · monthly rhythm · quarterly review · annual inscription (all concept suggestions)



Spring · developers' conference

Main venue Zhongzhiyuan forecourt; opens with the annual scenario list and a version K–marker review
KPI: attendance / projects converted / re–run appraisal

Summer · open–source camp

Permanent base at the Open Source House; member co–creation, mentor pairing, corporate challenges answered by the community
KPI: projects completed / talent–station residencies

Autumn · art and city festival

Whole axis, with arts institutions; the public open day follows the one–day experience route
KPI: public reach / perceived scenario quality

Winter · release season and honours night

Annual plaque unveiled at the Gradient Steps; honour signals lit
KPI: milestone selection quality / international reach

Routine rhythm and exit mechanism

Weekly · community day at the Open Source House   Monthly · scenario list review (switch meeting: pass / hold / reject)
Quarterly · scenario KPI review and yellow/red–card reappraisal   Annually · version K–marker review and event appraisal
Every event carries KPIs for attendance, talent and project conversion, and repeated underperformance triggers adjustment or exit; operating bodies and funding must be settled through statutory procedure. This calendar is a mechanism suggestion.

Every number recomputable — an auditable proposal

All known metrics are recomputed from geometry/*.geojson in EPSG:4548; displayed values match metrics.json

Overall design area
11.41 km^2
site_area_sqm · provisional

Green ratio
17.07 %
green_ratio

Public space ratio
11.76 %
public_space_ratio · incl. axis band

Concept floorspace
1.263 M m^2
proposed_total_floor_area_sqm

Slow–mobility network
36.6 km
road_network_length_m

Key areas total
369.3 ha
key_area_total_sqm

AI scenario node
12 nodes
scenario_node_count

Renewal projects
18 projects
renewal_project_count

Transcribed from official originals — competent–authority and official–system figures (not this proposal's control values)

Heritage park, whole line about 9 km, serving 9 Haidian subdistricts (planning commission, Dec 2021) · phase one 16.8 ha (official length given as both 2.4 and 2.5 km; this package keeps both)
Phase two 530,900 m^2, of which the northern section is 30.01 ha (both records describe phase two's ancillary project: landscape bureau, Jul 2026, ancillary works finished; public–resources platform performance notice, Aug 2026, records "completed: yes" — not an acceptance filing)
Seven planning plots in the belt carry approved plot ratios of 2.20–5.00 and height limits of 24–80 m (three original Planning Conditions) — other parties' approved values; this proposal sets no intensity from them

Evidence chain: from provisional boundary to self–check status

Provisional extent (not a red line) → Nine layers 39 cells → EPSG:4548 recompute → metrics.json 50 done · 4 pending → Three matrices 23 · 6 · 15 items → Self–check 4 gates passed

All derived from one source: figures · A3 booklet · A0 boards · visual/index.html (data–metric values match metrics.json)

Control values pending official data

Plot ratio · building height · building density · green ratio
(official_*_control in metrics.json)
To be supplied by: approved regulatory plan /
official task book annex
*All area metrics recompute once the red line arrives

Risk · Copyright · Missing Data

See chapter twelve of the proposal and risk.json

Principal risks

Spatial dispute (4): provisional vs real boundary deviation
Policy uncertainty (4): regulatory and renewal policy undecided
Implementation complexity (4): many parties along the corridors
Technology maturity (3): uncertainty in test scenarios

Mitigation and review

Provisional status marked package–wide; recomputation checklist locked
All policy content written as research suggestion
Three–phase rolling delivery: demonstrate before scaling
Test scenarios never enter regular operation without permission

Copyright statement

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Name and logo are concept suggestions pending trademark search
This platform is not the organiser's official entry channel

Data to be supplied

Official red line and key–area polygons
Approved regulatory conditions (plot ratio, height, density, green)
Existing buildings and tenure · heritage protection lines
Municipal capacity and traffic model data

Sources and Rights Summary

Formal and background sources kept separate · rights ledger independently verifiable

Formal sources (approved list)

Announcement · task book · site package
Public source registry · processed fact pack · provisional boundary
MOHURD urban design and regulatory plan measures ·
MNR land–use classification guidelines
→ Formal facts and normative arguments come only from this list

Background sources (no formal weight)

OSM existing features (ODbL 1.0, with NOTICE and
in–package archive visual/assets/osm–context.json)
Eight global case entry points · public industry reporting
The open–call launch article
→ Context and display only; never red lines, measurement or commitments

Rights ledger (independently verifiable)

Fonts: both PDFs measured at zero /FontFile entries;
text is Type 3 vector outline with no installable font embedded
(verify by searching for /FontFile and /Type3)
Logo: original in–package vector, unregistered, no trademark claim
Dependencies: open–source libraries used to generate only

Accessibility and licence self–check

Measured contrast: body 12.7:1 / brand green 4.8:1 / muted 5.6:1
SVG carries role and aria–label alternatives; no keyboard traps
Layered licensing: original content CC–BY–4.0; the OSM layer is
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Full ledger: report/copyright_statement.md